Table 1: Performance of the hybrid sentiment classifier under three train-test approaches

	China test data					US test data				
A. Three classes										
	F1			Acc.		F1			Acc.	
	Neg.	Amb.	Pos.	(%)	n	Neg.	Amb.	Pos.	(%)	n
Approach 1	0.90	0.62	0.75	75.9	83	0.44	0.46	0.41	44.6	130
Approach 2	0.79	0.40	0.76	69.9	73	0.84	0.39	0.63	68.6	70
Approach 3	0.76	0.63	0.68	68.4	95	0.58	0.62	0.47	57.9	107
B. One-vs-rest AUC and positive/negative accuracy										
	AUC			Acc.		AUC			Acc.	
	Neg.	Amb.	Pos.	(%)	n	Neg.	Amb.	Pos.	(%)	n
Approach 1	0.96	0.56	0.91	91.4	58	0.64	0.50	0.75	80.0	55
Approach 2	0.88	0.49	0.89	87.9	58	0.88	0.62	0.89	85.5	55
Approach 3	0.96	0.56	0.87	92.9	56	0.74	0.54	0.80	78.0	50

Notes: The hybrid score averages the SVM label, the random forest label and the standardized Bing Liu lexicon score. Approach 1 trains on positive and negative documents only; approach 2 adds a sample of 75 ambiguous documents; approach 3 trains separate positive-vs-not and negative-vs-not models and combines them. Panel A: the continuous hybrid score is cut into three classes at quantiles matching the share of each hand-coded class in the evaluation set; for approach 1, which has no ambiguous documents in its test split, held-out ambiguous documents are added to the evaluation set. Panel B: AUC treats each class in turn as the outcome and the hybrid score as the predictor, with the direction of the score chosen for each class (the pROC default); accuracy classifies hand-coded positive and negative documents by the sign of the hybrid score. n is the number of evaluation documents.